

Qatar-1b

R-2686

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120609 · created 2026-08-31T00:35:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456087.84085 +/- 0.0002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.024
Transit depth (Rp/Rs)^2	2.99 +/- 0.84 [%]
Orbital Inclination (inc)	81.93 +/- 1.46
Airmass coefficient 1 (a1)	1.067 +/- 0.011
Airmass coefficient 2 (a2)	-0.052 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	None
Optimal Aperture	3.47
Optimal Annulus	12.11
Transit Duration (day)	0.054 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.024 vs 0.1463 (deviation 1.07σ)
Duration fit vs published	0.054 d vs 0.0692 d (deviation 0.91σ)
Mid-time O–C	0.4 min
Depth SNR	6.9
Residual scatter	1.07 %

Light curve

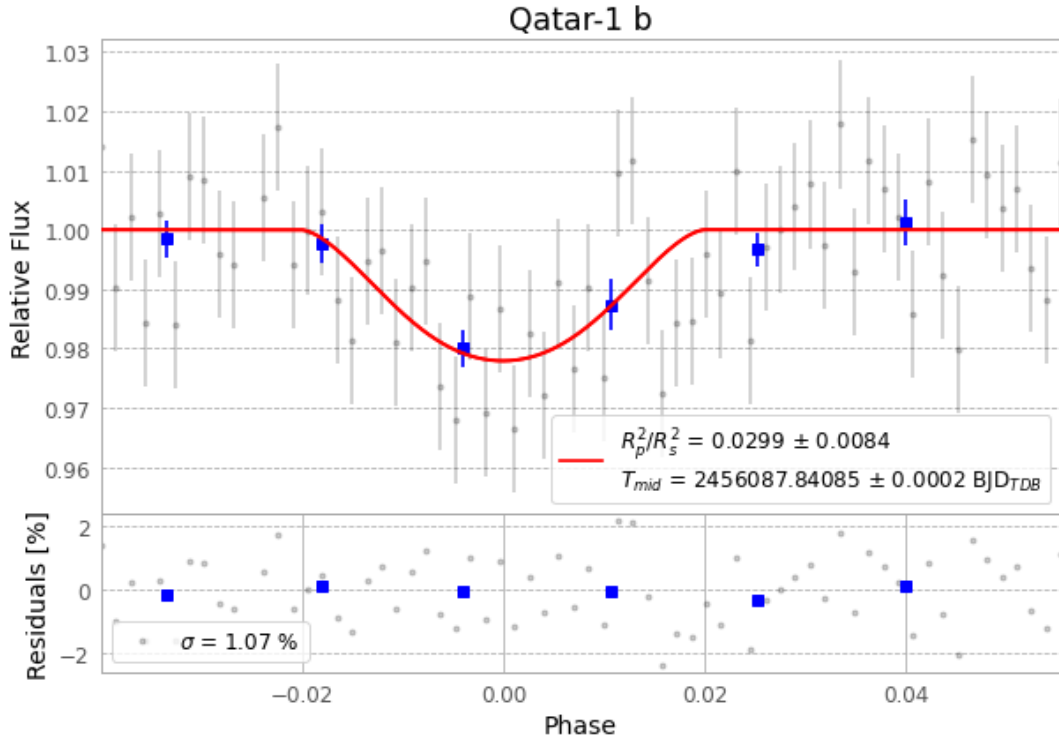


Figure 1 — FinalLightCurve_Qatar-1 b_2012-06-08.png (SHA-256 9ec973a678a33628...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [424, 211], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 7daafd1625f7b4df...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), Qatar-1120609064812.FITS → Qatar-1120609100313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120609.log (4f46ba60404d...), FinalLightCurve_Qatar-1 b_2012-06-08.png (9ec973a678a3...), FinalLightCurve_Qatar-1 b_2012-06-08.csv (5a711845fcad...)

Compute resources

Wall time 204.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 00:35Z.